



Exercise Session 6 Highlights

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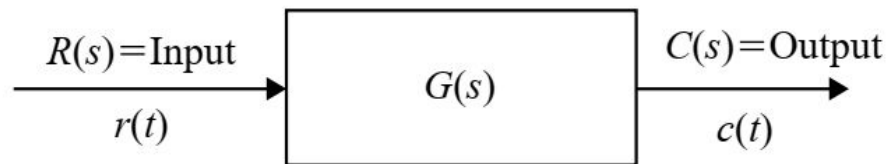
Course: Reglerteknik (EL1000)

2023

Frequency Response

The frequency response of a system is defined as the steady-state response of the system to a sinusoidal input signal. The sinusoid is a unique input signal, and the resulting output signal for a linear system is sinusoidal in the steady state; it differs from the input only in amplitude and phase angle.

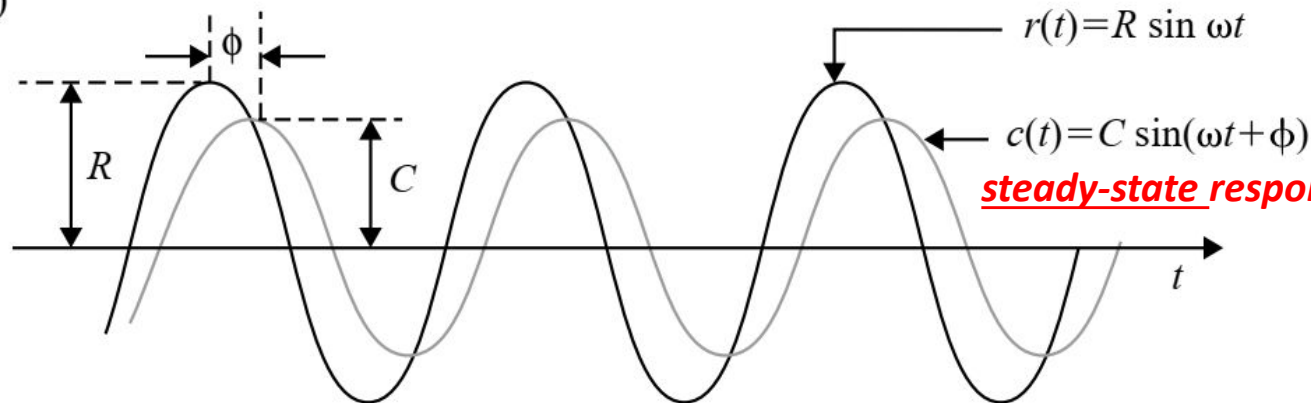
(a)



$$\frac{C}{R}(j\omega) = G(j\omega)$$

Sinusoidal transfer function ($s = j\omega$)

(b)



steady-state response : $\lim_{t \rightarrow \infty} C(t)$

(transient Response has died out!)

In general, one can show that (for stable $G(s)$):

$$r(t) = A \cos \omega t + B \sin \omega t = \sqrt{A^2 + B^2} \cos [\omega t - \tan^{-1}(B/A)] \Rightarrow r(t) = M_i \cos(\omega t + \phi_i)$$

$$G(j\omega) = |G(j\omega)| \angle G(j\omega) = M(\omega) \angle \phi(\omega)$$

$$C(t)_{t \rightarrow \infty} = M_o \cos(\omega t + \phi_o) = M_i M(\omega) \cos(\omega t + \phi_o), \quad \phi_o = \phi_i + \phi$$

Plotting Frequency Response: 1) Polar Plot (incomplete Nyquist plot)

The frequency response is characterized by the **magnitude** and **phase** of the sinusoidal transfer function $G(j\omega)$

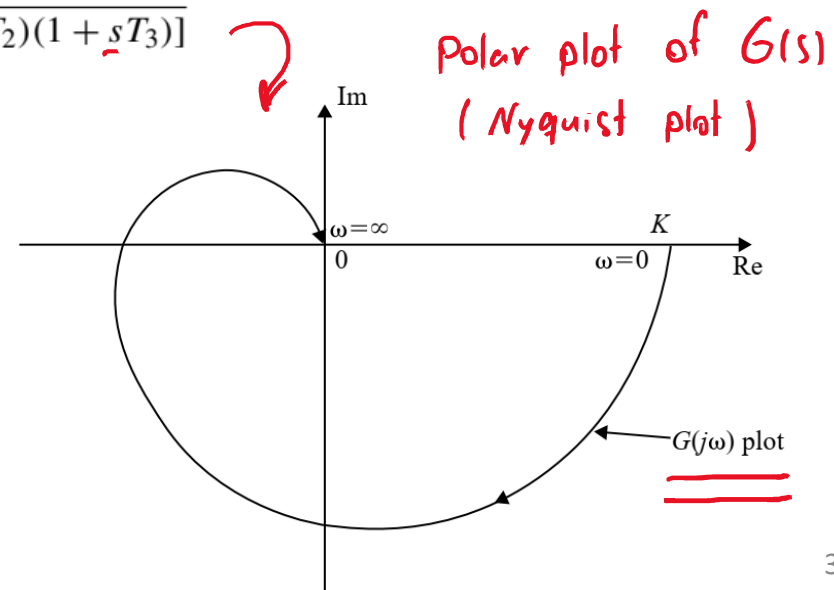
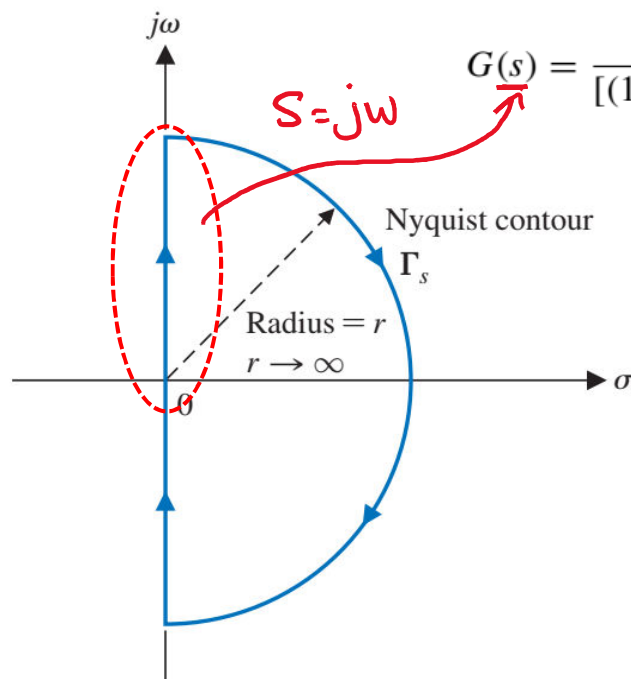
The transfer function of a system $G(s)$ can be described in the frequency domain by the relation:

$$G(j\omega) = G(s) \big|_{s=j\omega} = R(\omega) + jX(\omega)$$

Alternatively, the transfer function can be represented by

$$G(j\omega) = |G(j\omega)| e^{j\phi(\omega)} = |G(j\omega)| \angle \phi(\omega)$$

The polar plot of $G(s)$ is the plot of the magnitude of $G(j\omega)$ versus the phase angle of $G(j\omega)$ (on the complex plane) as ω is varied from zero to infinity



Plotting Frequency Response: 2) Bode Plots

$$G(s) = 1/(s + 2)$$

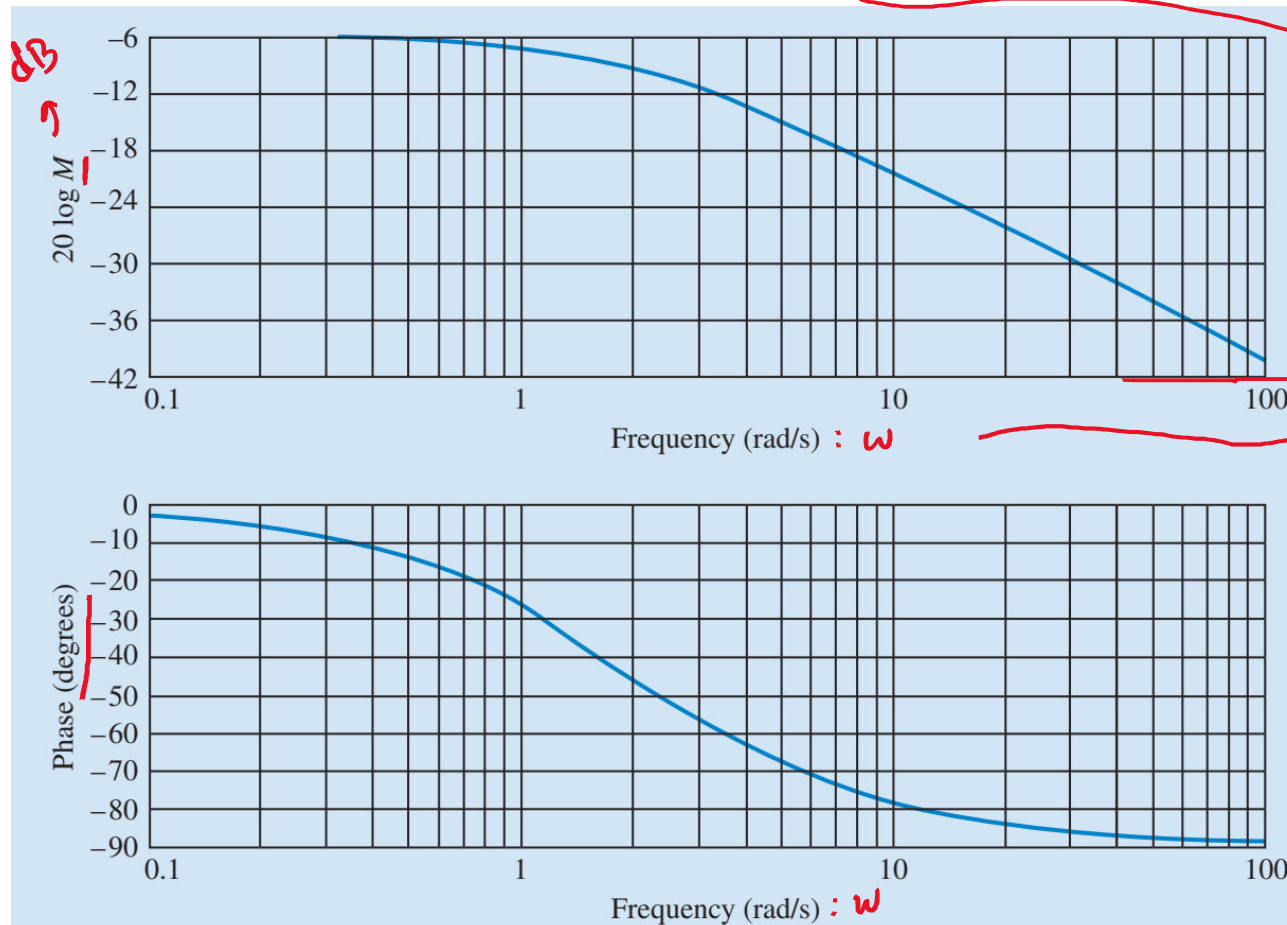
$$G(j\omega) = 1/(j\omega + 2) = (2 - j\omega)/(\omega^2 + 4)$$

$$\text{Logarithmic gain} = 20 \log_{10}|G(j\omega)|,$$

where the units are **decibels (dB)**

$$|G(j\omega)| = M(\omega) = 1/\sqrt{(\omega^2 + 4)}$$

$$\phi(\omega) = -\tan^{-1}(\omega/2) \rightarrow \text{phase angle}$$



Bode Magnitude Plot

Example (almost a general transfer function):

$$G(s) = \frac{K(1 + sT_1)(1 + sT_2) \cdots (1 + sT_m)}{s^k(1 + sT_a)(1 + sT_b) \cdots (1 + sT_n)}$$

G(s) is in Time Constant Form

$$G(j\omega) = \frac{K(1 + j\omega T_1)(1 + j\omega T_2) \cdots (1 + j\omega T_m)}{(j\omega)^k(1 + j\omega T_a)(1 + j\omega T_b) \cdots (1 + j\omega T_n)}$$

Sinusoidal transfer function

$$\angle G(j\omega) = \boxed{-k \times 90^\circ} + \tan^{-1} \omega T_1 + \tan^{-1} \omega T_2 + \cdots + \tan^{-1} \omega T_m - (\tan^{-1} \omega T_a + \tan^{-1} \omega T_b + \cdots + \tan^{-1} \omega T_n)$$

} Phase angle ...

$$|G(j\omega)| = \frac{K \sqrt{(1 + \omega^2 T_1^2)(1 + \omega^2 T_2^2) \cdots (1 + \omega^2 T_m^2)}}{\omega^k \sqrt{(1 + \omega^2 T_a^2)(1 + \omega^2 T_b^2) \cdots (1 + \omega^2 T_n^2)}}$$

$20 \log |G(j\omega)| : \text{dB}$

it becomes
the summation
of different
factors of G(jw)
in dB

$$\text{dB} = 20 \log \frac{K}{\omega^k} + 20 \log \sqrt{(1 + \omega^2 T_1^2)} + \cdots + 20 \log \sqrt{(1 + \omega^2 T_m^2)} - 20 \log \sqrt{(1 + \omega^2 T_a^2)} - \cdots - 20 \log \sqrt{(1 + \omega^2 T_n^2)}$$

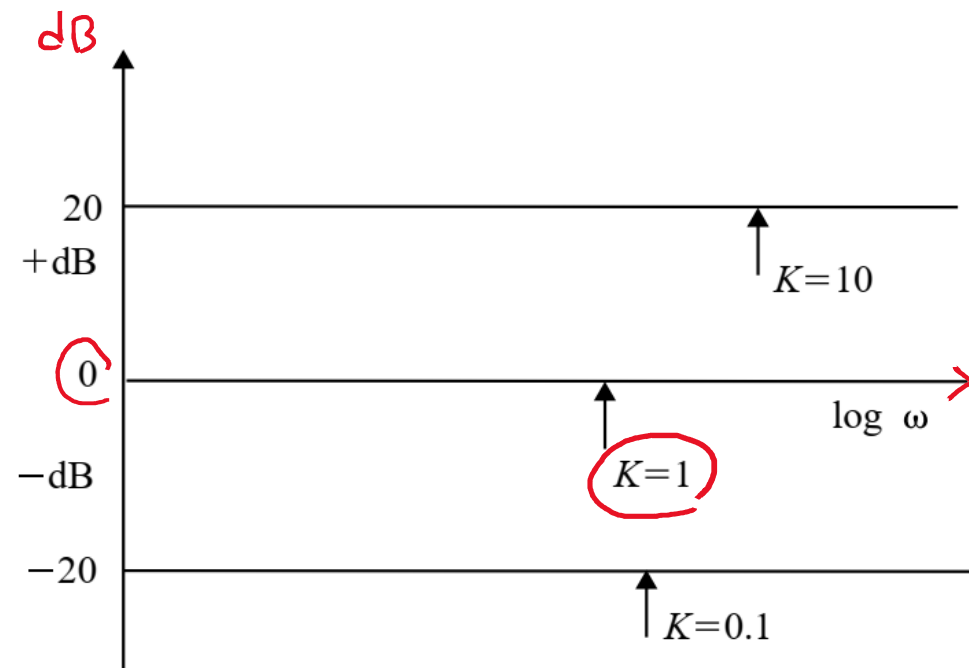
Bode Asymptotic Magnitude Plot

$G(j\omega)$ can contain just five simple **factors** which are given below:

1. $G(j\omega) = K$ (a constant factor)
 2. $G(j\omega) = K(j\omega)^{\pm p}$ (poles or zeros at the origin of order p)
 3. $G(j\omega) = (1 + j\omega T)^{\pm q}$ (poles or zeros at $s = -(1/T)$ of order q)
 4. $G(j\omega) = [\omega_n^2 / -\omega^2 + j2\zeta\omega_n\omega + \omega_n^2]^{\pm r}$ (complex poles and zeros of order r)
 5. $G(j\omega) = e^{-j\omega T^d}$ (Time delay)
-

1) A constant factor (gain)

$$\begin{aligned}G(s) &= K \\G(j\omega) &= K \\|G(j\omega)| &= K \\dB &= 20 \log K\end{aligned}$$



Bode Asymptotic Magnitude Plot (cont.)

$b + mn = y$: line equation

2) Zeros/Poles at the origin

$$G(s) = Ks$$

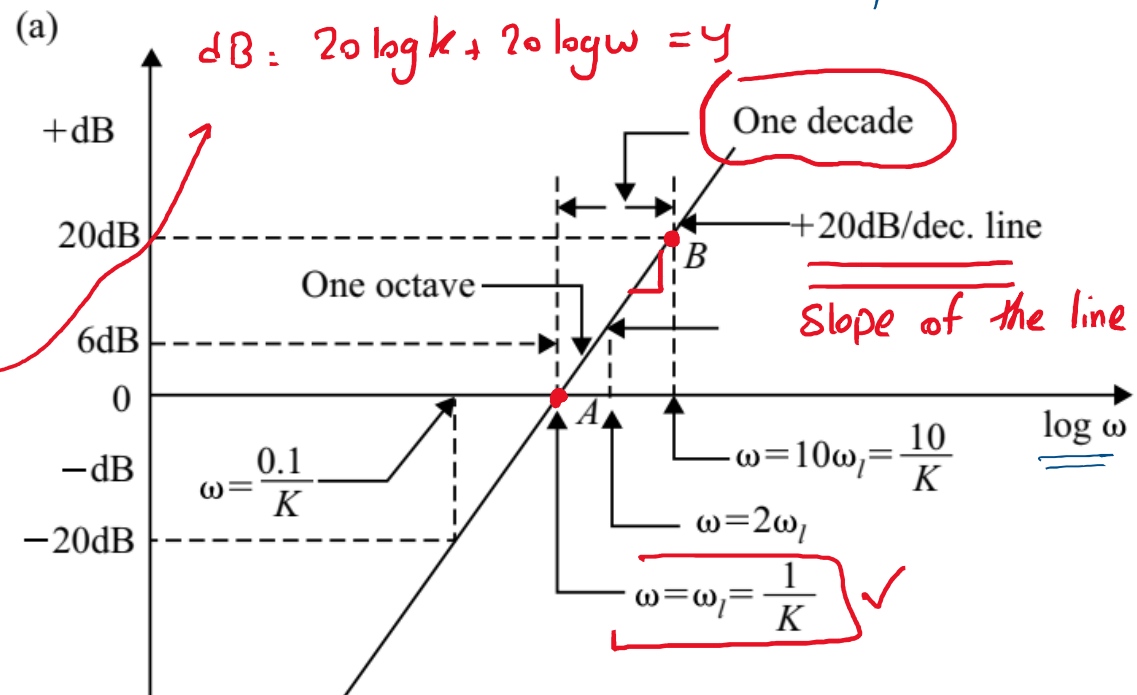
$$G(j\omega) = jK\omega$$

$$y = |G(j\omega)| = K\omega$$

$$\text{dB} = 20 \log K\omega$$

$$\text{At } \omega = \frac{1}{K}, \quad \text{dB} = 20 \log 1 = 0$$

$$\text{At } \omega = \frac{10}{K}, \quad \text{dB} = 20 \log 10 = 20$$



The meaning of **decade** is the frequency range which is multiplied or divided by the factor 10.

If the frequency 1 is multiplied by 10, the frequency range from 1 to 10 is a decade.

The frequency range from 10 to 100 is also a decade.

The frequency range from 100 to 1000 is also a decade.

The frequency range from 1 to 100 is two decades ...

The frequency range from 1 to 1000 is 3 decades ...

$$\log \frac{\omega_2}{\omega_1}$$

gives the freq. range
from ω_1 to ω_2 in decade

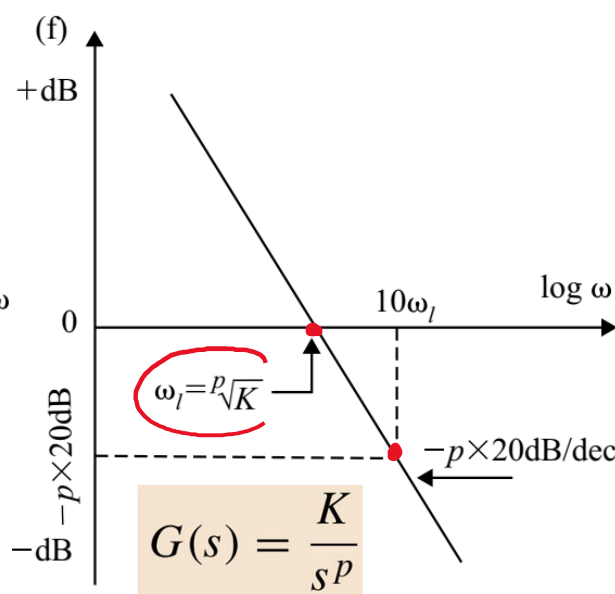
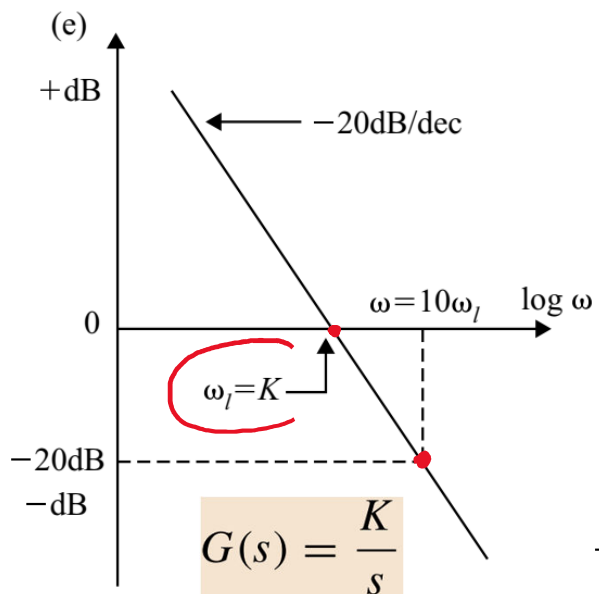
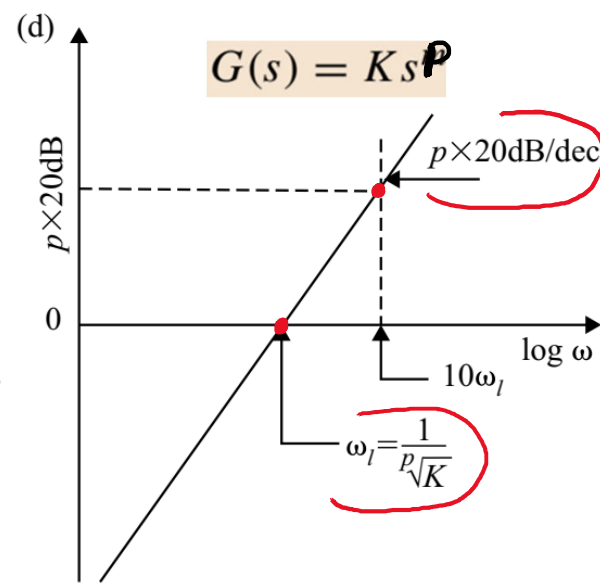
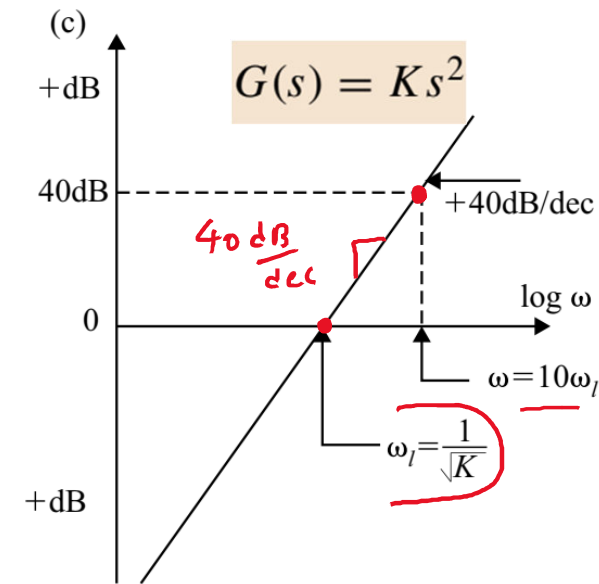
ex. $\omega_1 = 1$, $\omega_2 = 1000$

$$\log \frac{\omega_2}{\omega_1} = 3 \text{ decade}$$

Octave means, the frequency is multiplied or divided by a factor of 2

Bode Asymptotic Magnitude Plot (cont.)

2) Zeros/Poles at the origin (cont.)



(c)

$$G(s) = Ks^2$$

$$|G(j\omega)| = K\omega^2$$

$$K\omega_l^2 = 1$$

$$\omega_l = \frac{1}{\sqrt{K}}$$

$$\omega = 10\omega_l, \text{ dB is } 20 \log K(10/\sqrt{K})^2 = 40 \text{ dB}$$

(e)

$$G(s) = \frac{K}{s}$$

$$|G(j\omega)| = \frac{K}{\omega}$$

$$|G(j\omega)| = 1 \text{ at } \omega = K \text{ or } \omega_l = K$$

$$\omega = 10\omega_l = 10K$$

$$\text{dB} = 20 \log \frac{K}{10K} = -20$$

Bode Asymptotic Magnitude Plot (cont.)

3) Zeros/Poles at $s = - (1/T)$

$$G(s) = (1 + sT)$$

$$G(j\omega) = 1 + j\omega T$$

$$|G(j\omega)| = \sqrt{(1 + \omega^2 T^2)}$$

for $\omega T \ll 1$ $\omega \ll \frac{1}{T}$

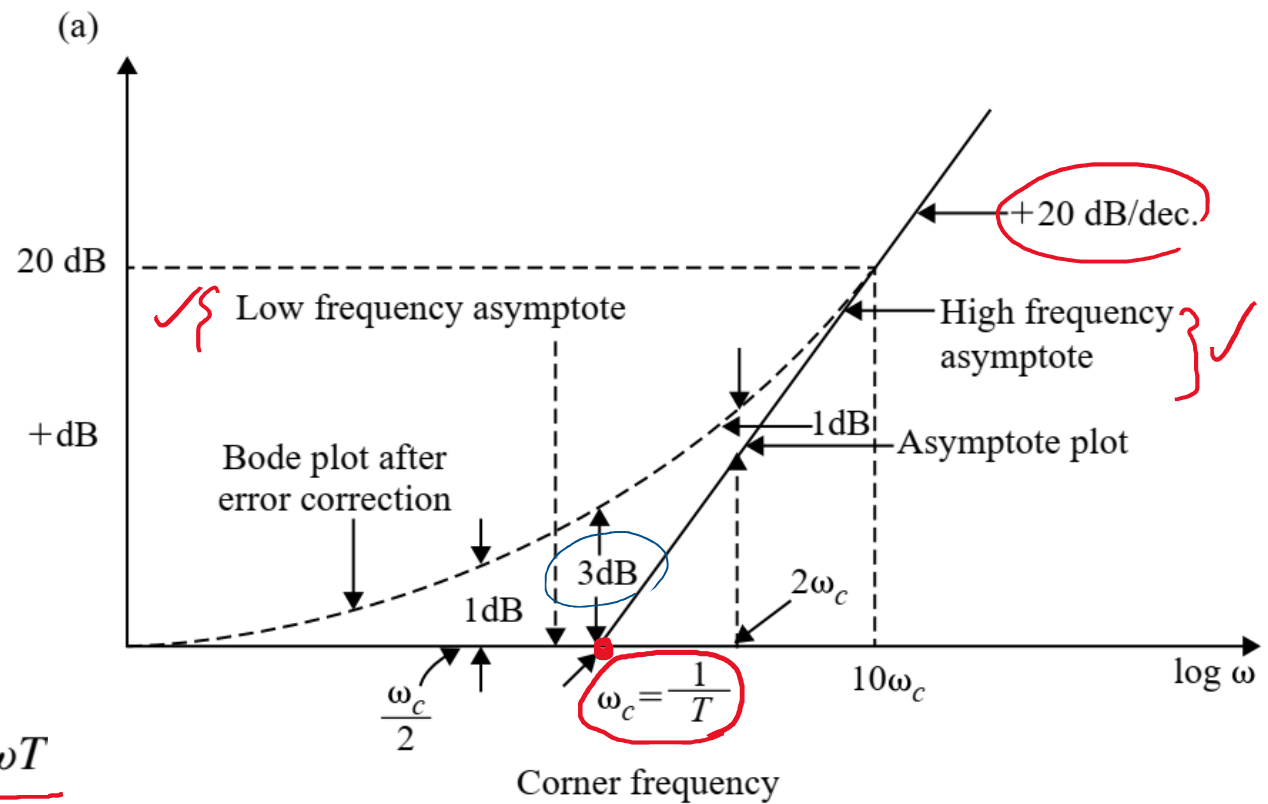
$$|G(j\omega)| = 1$$

$$\text{dB} = 20 \log 1 = 0$$

For $\omega T \gg 1$ $\omega \gg \frac{1}{T}$ $|G(j\omega)| = \omega T$

the same as having a zero at the origin : $G(s) = Ts$

* Asymptotes give approximate behavior around $\omega = \frac{1}{T}$



The **corner frequency** of a Bode plot is defined as the frequency at which the two asymptotes (approximate plots) meet.

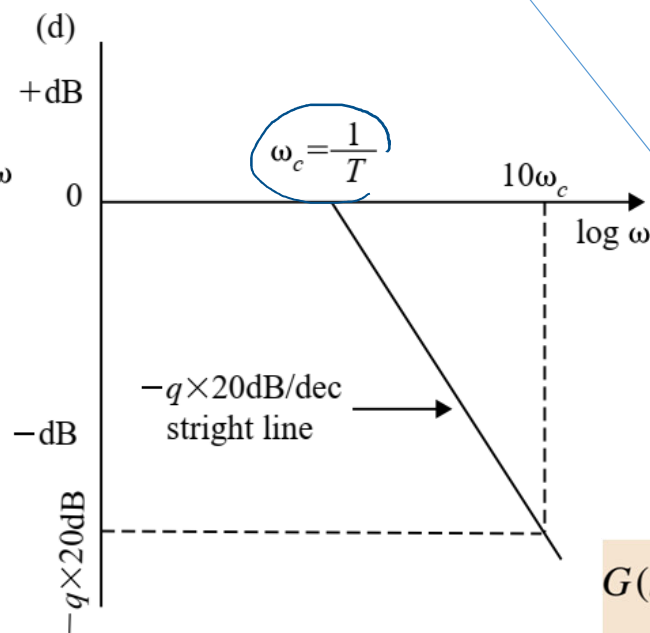
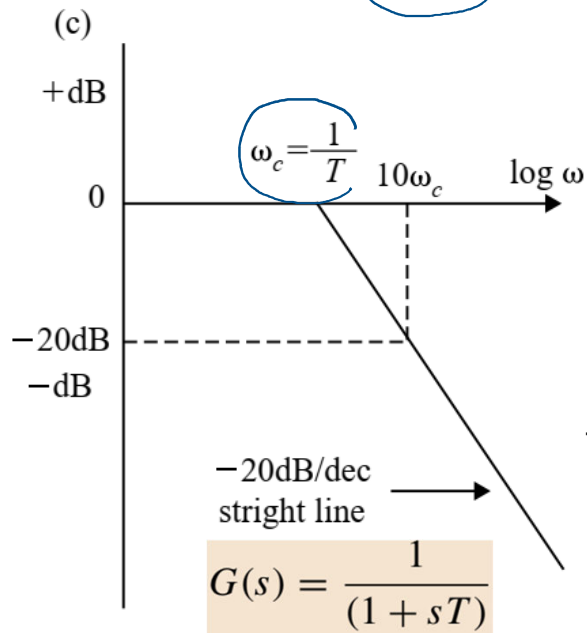
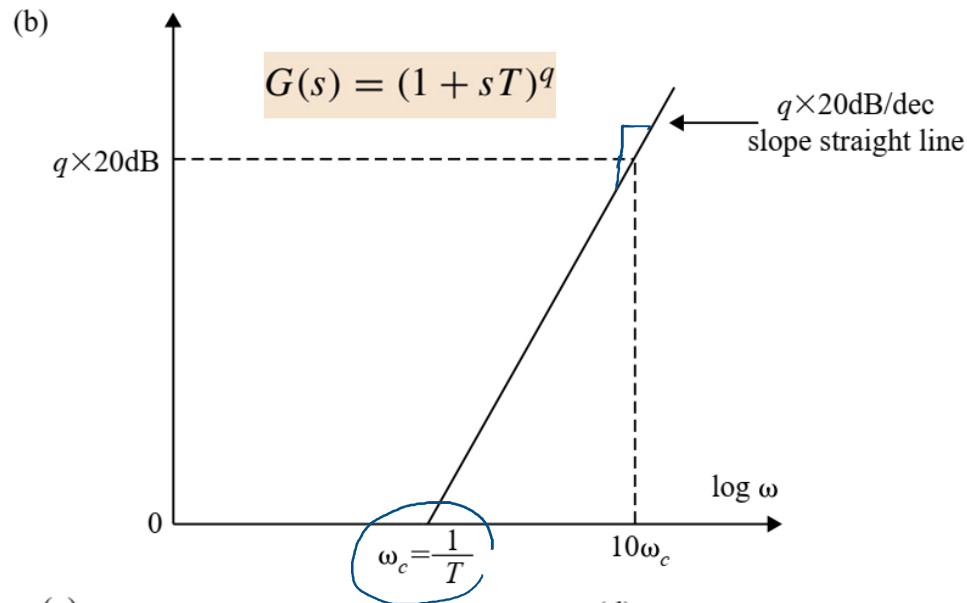
Correction at the Corner Frequency

$$|G(j\omega)| = \sqrt{1 + \omega^2 T^2}$$

$$\text{dB}|_{\omega=\frac{1}{T}} = 20 \log \sqrt{2} = \underline{\underline{3 \text{ dB}}}$$

Bode Asymptotic Magnitude Plot (cont.)

3) Zeros/Poles at $s = -(1/T)$ cont.



(b)

$$G(s) = (1 + sT)^q$$

$$G(j\omega) = (1 + j\omega T)^q$$

$$|G(j\omega)| = [\sqrt{(1 + \omega^2 T^2)}]^q$$

$$\text{dB} = 10 \times q \log(1 + \omega^2 T^2)$$

$$\omega^2 T^2 \gg 1 : \text{dB} = 10q \log \omega^2 T^2$$

$$(\omega \gg \frac{1}{T})$$

$$= 20q \log \omega T$$

$$= 20q (\log T + \log \omega)$$

high freq.
asymptote

$$\omega^2 T^2 \ll 1 :$$

$$(\omega \ll \frac{1}{T})$$

$$\text{dB} = 10q \log 1 = 0$$

low freq. asymptote

Bode Asymptotic Magnitude Plot (cont.)

4) Complex poles

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

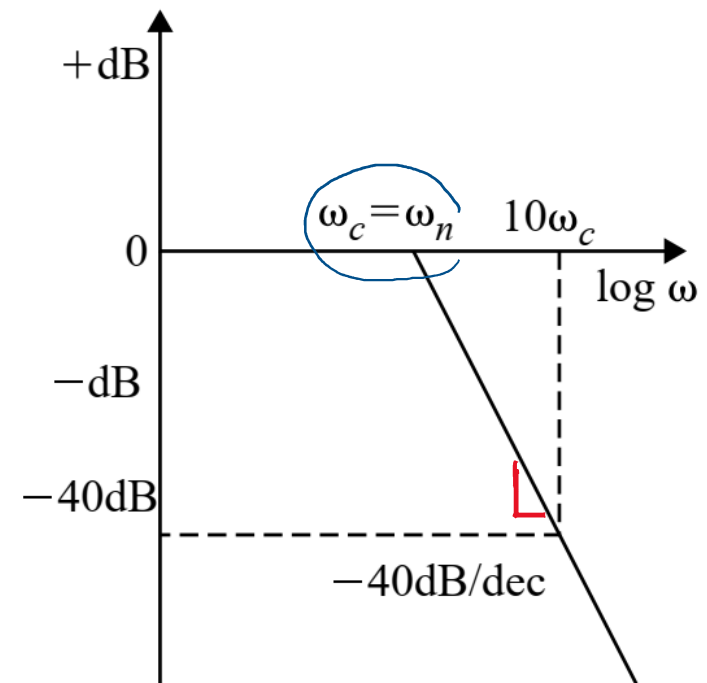
$$G(j\omega) = \frac{\omega_n^2}{-\omega^2 + j2\zeta\omega\omega_n + \omega_n^2}$$

$$\begin{aligned} |G(j\omega)| &= \frac{\omega_n^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega^2\omega_n^2}} \\ &= \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + 4\zeta^2\frac{\omega^2}{\omega_n^2}}} \end{aligned}$$

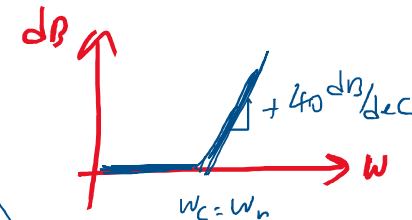
For at $\omega \ll \omega_n$, $|G(j\omega)| = 1$
dB = 0

For $\omega \gg \omega_n$, $|G(j\omega)| = \frac{\omega_n^2}{\omega^2}$

$$\begin{aligned} \text{dB} &= 20 \log \frac{\omega_n^2}{\omega^2} \\ &= -40\text{dB/dec with } \omega_c = \omega_n \end{aligned}$$



for complex zeros
 $G(s) = \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$



Bode Asymptotic Magnitude Plot (Summary)

T.F. $G(s)$	K	$\frac{K}{s}$	$\frac{K}{s^2}$	Ks	Ks^2	$(1 + sT)$
Corner frequency, ω_c	—	K	$\sqrt{K}$	$\frac{1}{K}$	$\frac{1}{\sqrt{K}}$	$\frac{1}{T}$
Slope, dB/decade	0	−20	−40	20	40	20
T.F. $G(s)$	$(1 + sT)^2$	$\frac{1}{(1 + sT)}$	$\frac{1}{(1 + sT)^2}$	$\frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)}$		
Corner frequency, ω_c	$\frac{1}{T}$	$\frac{1}{T}$	$\frac{1}{T}$	ω_n		
Slope, dB/decade	40	−20	−40	−40		

Note Phase plots are normally drawn for the actual phase angle which is calculated at each ω . The asymptotic phase plot is usually avoided

Bode Asymptotic Magnitude Plot Example

Consider the following T.F.

$$G(s) = \frac{4(s+5)(s+10)}{s^2(s+20)}$$

Sketch the Bode magnitude plot.

$$\begin{aligned} G(s) &= \frac{4(s+5)(s+10)}{s^2(s+20)} = \frac{10 \times 5 \times 4}{20} \frac{(1+\frac{1}{5}s)(1+\frac{1}{10}s)}{s^2(1+\frac{1}{20}s)} = \frac{10(1+\frac{1}{5}s)(1+\frac{1}{10}s)}{s^2(1+\frac{1}{20}s)} \\ &= \underbrace{\frac{10}{s^2}}_{G_1} \underbrace{(1+\frac{1}{5}s)}_{G_2} \underbrace{(1+\frac{1}{10}s)}_{G_3} \underbrace{\frac{1}{(1+\frac{1}{20}s)}}_{G_4} \end{aligned}$$

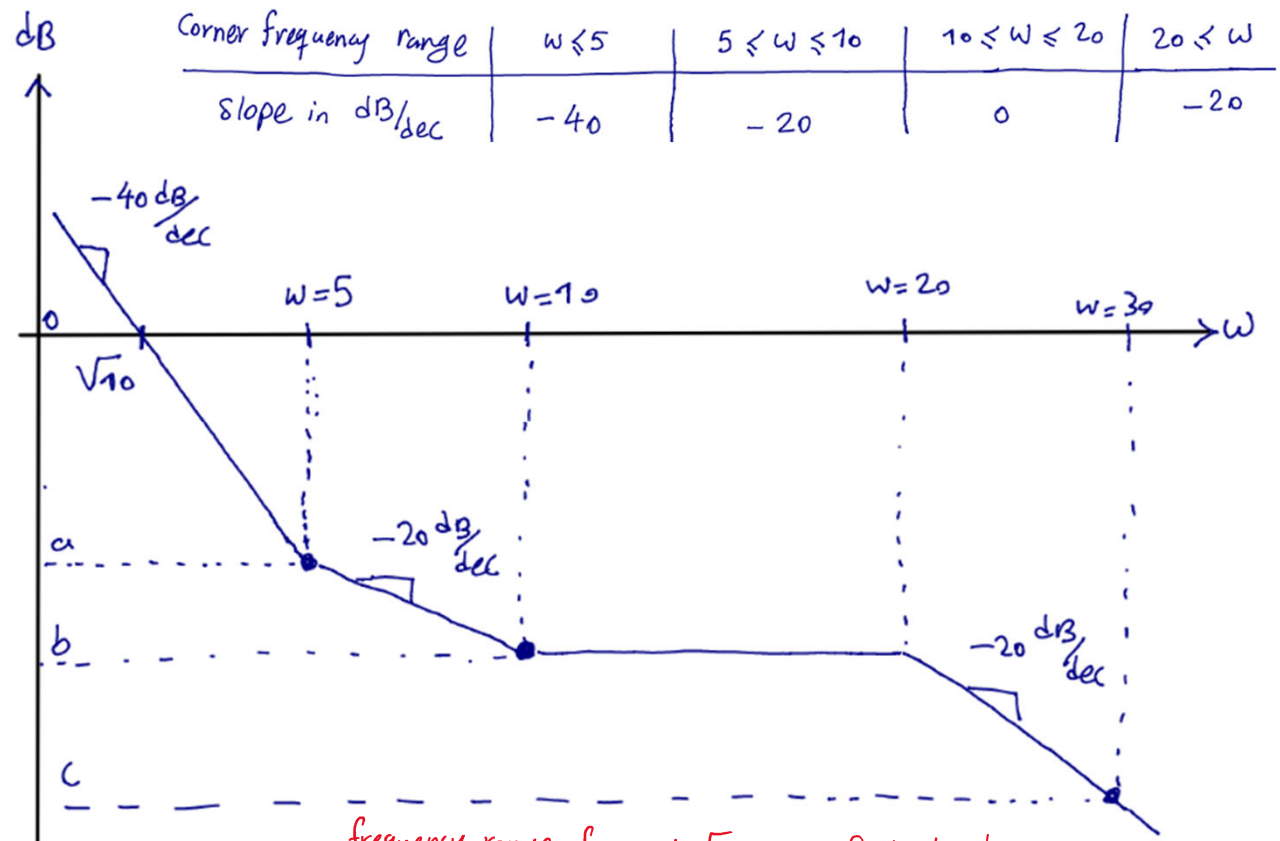
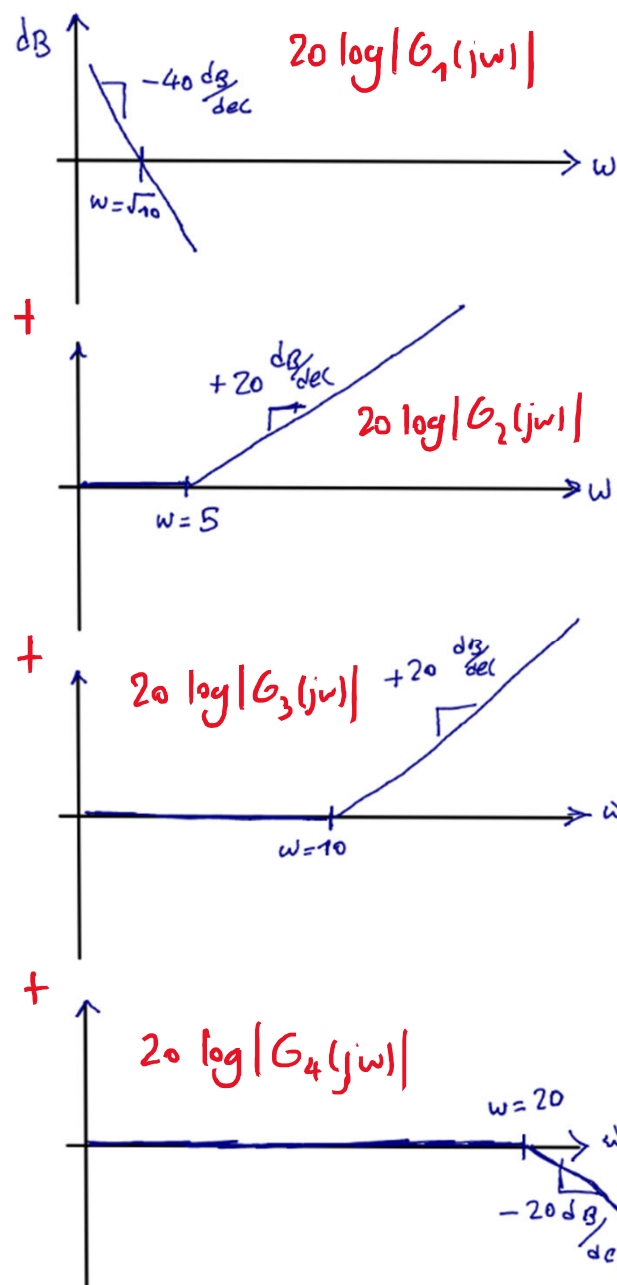
Transfer function in time constant form.

Factorizing the transfer function

$w_{c1} = \sqrt{10} \quad w_{c2} = 5 \quad w_{c3} = 10 \quad w_{c4} = 20$

$$20 \log |G(j\omega)| = 20 \log |G_1| + 20 \log |G_2| + 20 \log |G_3| + 20 \log |G_4|$$

Bode Asymptotic Magnitude Plot Example (cont.)



frequency range from $\omega = \sqrt{10}$ to $\omega = 5$ in decade

$$a - 0 = \underbrace{-40}_{\text{slope}} \log \frac{5}{\sqrt{10}} = -20 \log \frac{25}{10} \approx -8 \text{ dB}$$

$$b - a = -20 \log \frac{10}{5} \approx -6 \Rightarrow b \approx -14 \text{ dB}$$

$$c - b = -20 \log \frac{30}{20} \approx -3.5 \Rightarrow c \approx -17.5 \text{ dB}$$

Bode Asymptotic Magnitude Plot Example (cont.)

actual magnitudes in dB:

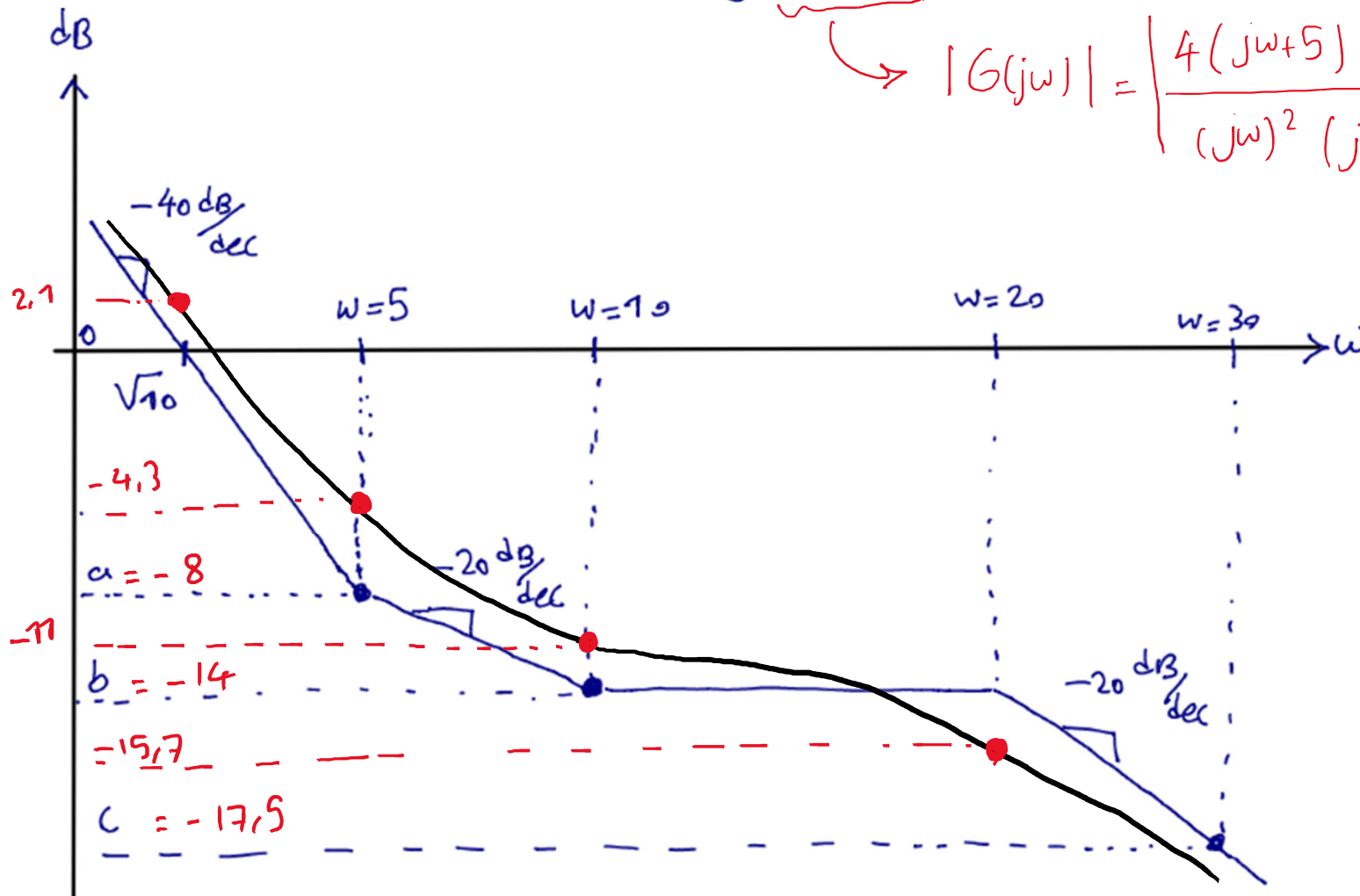
$$20 \log |G(j\sqrt{10})| \simeq 2,1$$

$$20 \log |G(j20)| \simeq -15,7$$

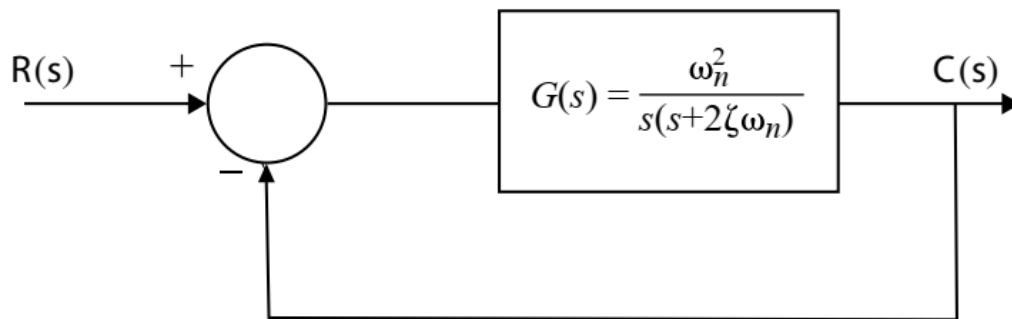
$$20 \log |G(j5)| \simeq -4,3$$

$$20 \log |G(j10)| \simeq -11$$

$$|G(j\omega)| = \left| \frac{4(j\omega+5)(j\omega+10)}{(j\omega)^2(j\omega+20)} \right|$$



Correlation Between Transient Response and Frequency Response



Prototype second order system

$$\frac{C(s)}{R(s)} = M(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$s_{1,2} = \underbrace{-\zeta\omega_n}_{\text{Real}} \pm \underbrace{\omega_n\sqrt{\zeta^2 - 1}}_{\text{imaginary } (\zeta < 1)}$$

The frequency at which the resonant peak occurs is called **Resonant Frequency** denoted by ω_r

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2}$$

ω_r should be real

Resonant peak
(maximum of bode magnitude plot):

$$M_r = \frac{1}{2\zeta\sqrt{1 - \zeta^2}}$$

Valid for:

$$0 \leq \zeta \leq 0.707$$

For $\zeta > 0.707$ there won't be any resonance.
Therefore: $\omega_r = 0$ and it can be shown that $M_r = 1$

Recall that: larger damping ratio $\zeta \Rightarrow$ smaller overshoot

ζ, ω_n characterize
transient response
behavior

See Ex Session 2 highlights ...

Correlation Between Transient Response and Frequency Response (Cont.)

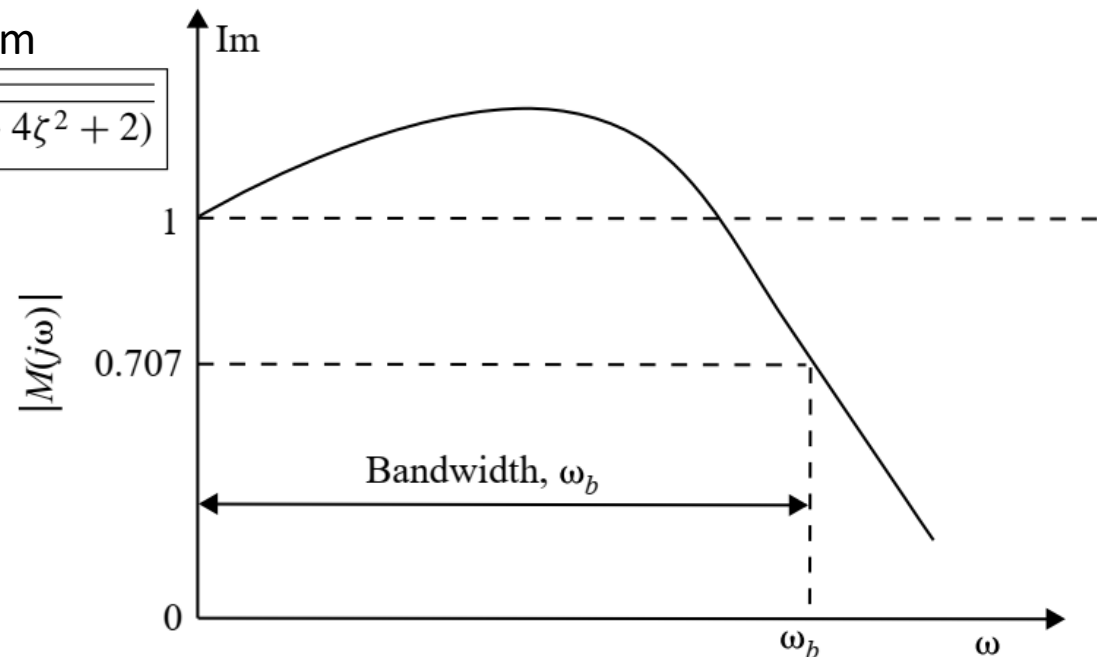
The **bandwidth** ω_b of the **closed loop frequency response** is defined as the frequency at which the magnitude response curve $|M(j\omega)|$ does not fall below 0.707 of its value at zero frequency.

In terms of decibel (dB), it is defined as the frequency at which $20\log_{10}|M(j\omega)|$ does not fall below -3 dB from its value at zero frequency.

For standard second order system

$$\text{Bandwidth} = \omega_b = \omega_n \sqrt{1 - 2\zeta^2 + \sqrt{(4\zeta^4 - 4\zeta^2 + 2)}}$$

Large bandwidth means the system will pass a wide range of frequencies of the input signal. In the time domain, it amounts to a smaller rise time, hence, faster transient response. Bandwidth is inversely proportional to the rise time.



Frequency response and bandwidth of a second order system

Gain margin (using Nyquist/Polar plot)

Let $G(s)$ be a **minimum phase** (open-loop) transfer function (i.e., $G(s)$ has no zeros and poles in the RHP).
 → Moreover for minimum phase open-loop T.F.s the Nyquist plot always get closed on the Right half plane. That's why the following can be claimed only using the

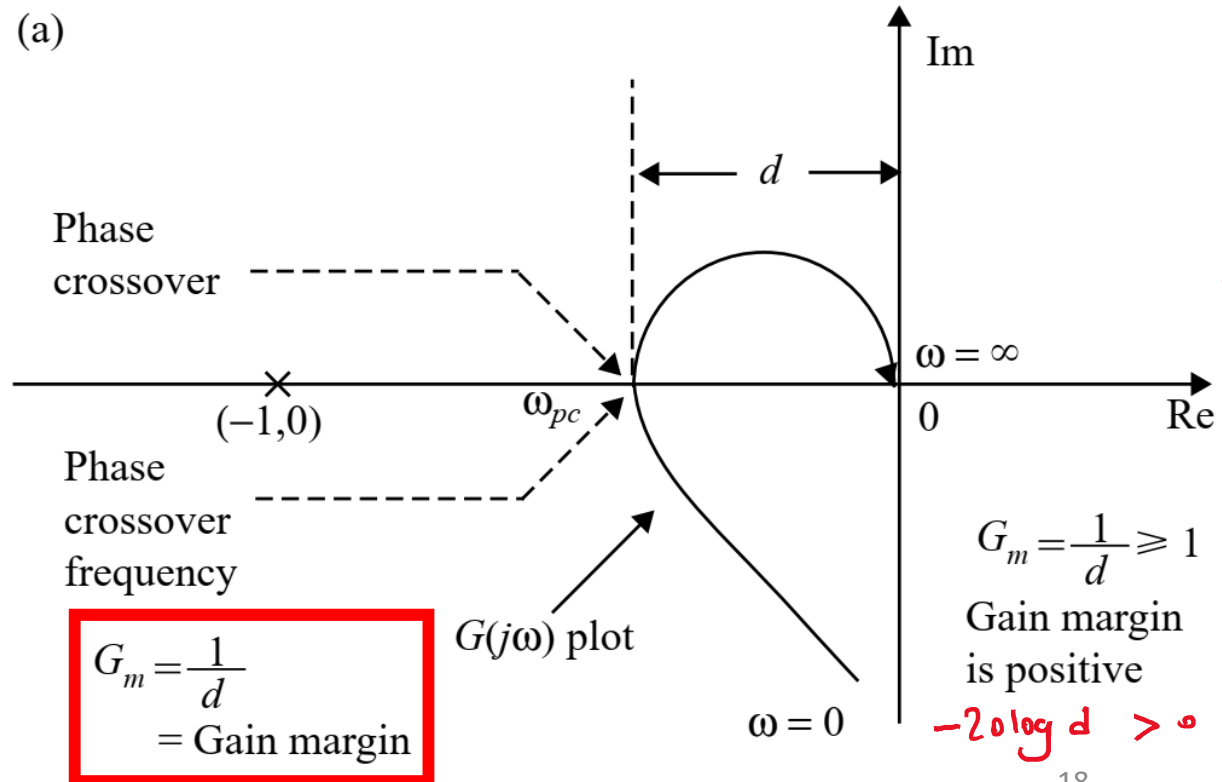
Consider the polar (incomplete Nyquist) plot of $G(j\omega)$: $Z_r = P_r + N_r$ polar plot $G(j\omega)$!
 Since $G(s)$ is a minimum phase T.F., in the Nyquist criterion we have $P_r = 0$.
 Therefore, for the closed-loop system stability it is sufficient that $G(j\omega)$ does not enclose $(-1, 0)$, i.e., $N_r = 0$, hence $Z_n = 0 + 0$ (no closed loop poles in RHP).

Phase crossover frequency ω_{pc} is defined as the frequency where the phase angle of $G(j\omega)$ is 180° .

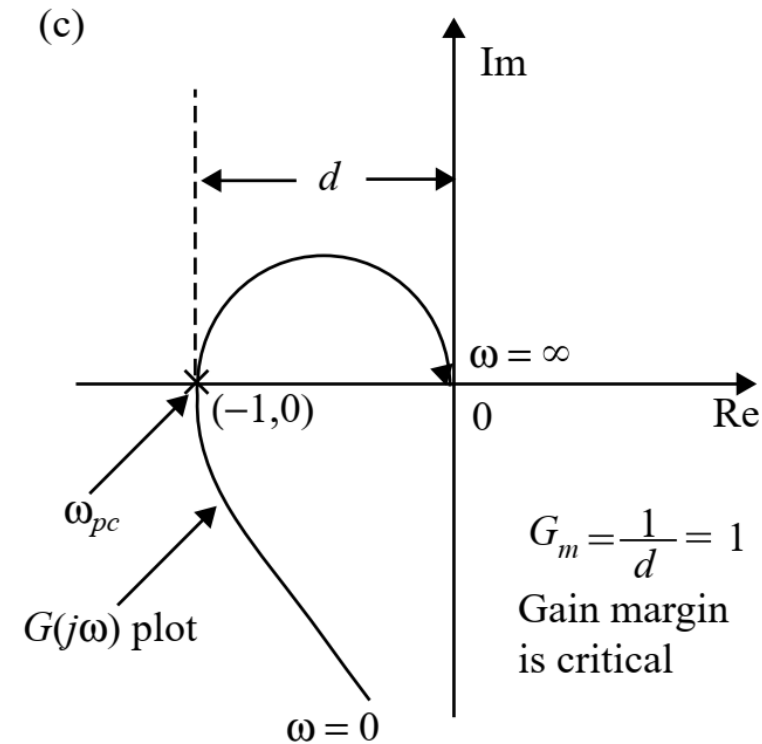
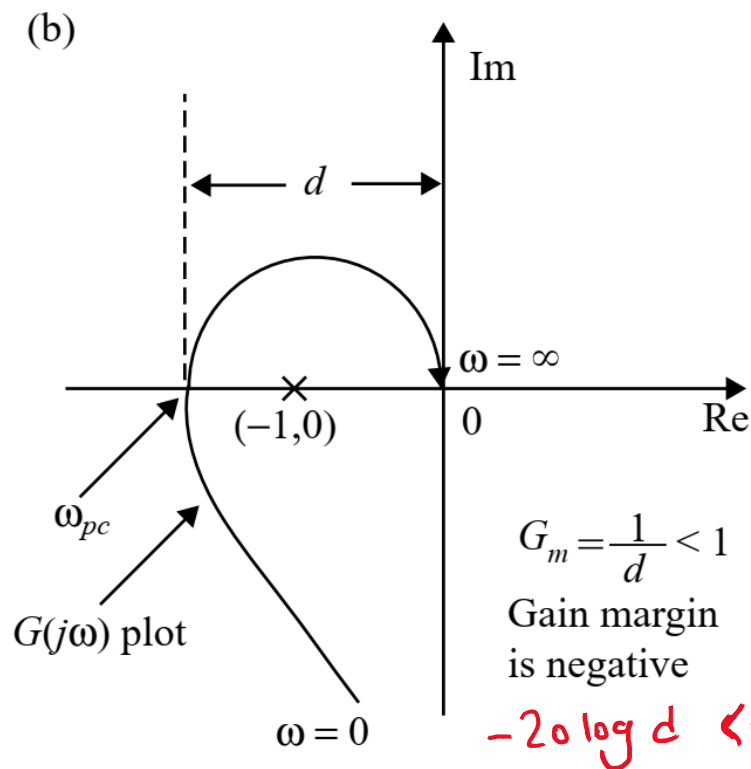
Gain margin G_m is defined as the factor (gain) by which the open-loop system gain can be increased before the closed-loop system becomes unstable.

Gain margin in dB:

$$G_m = -20 \log d$$



Gain margin (using Nyquist/Polar plot) cont.

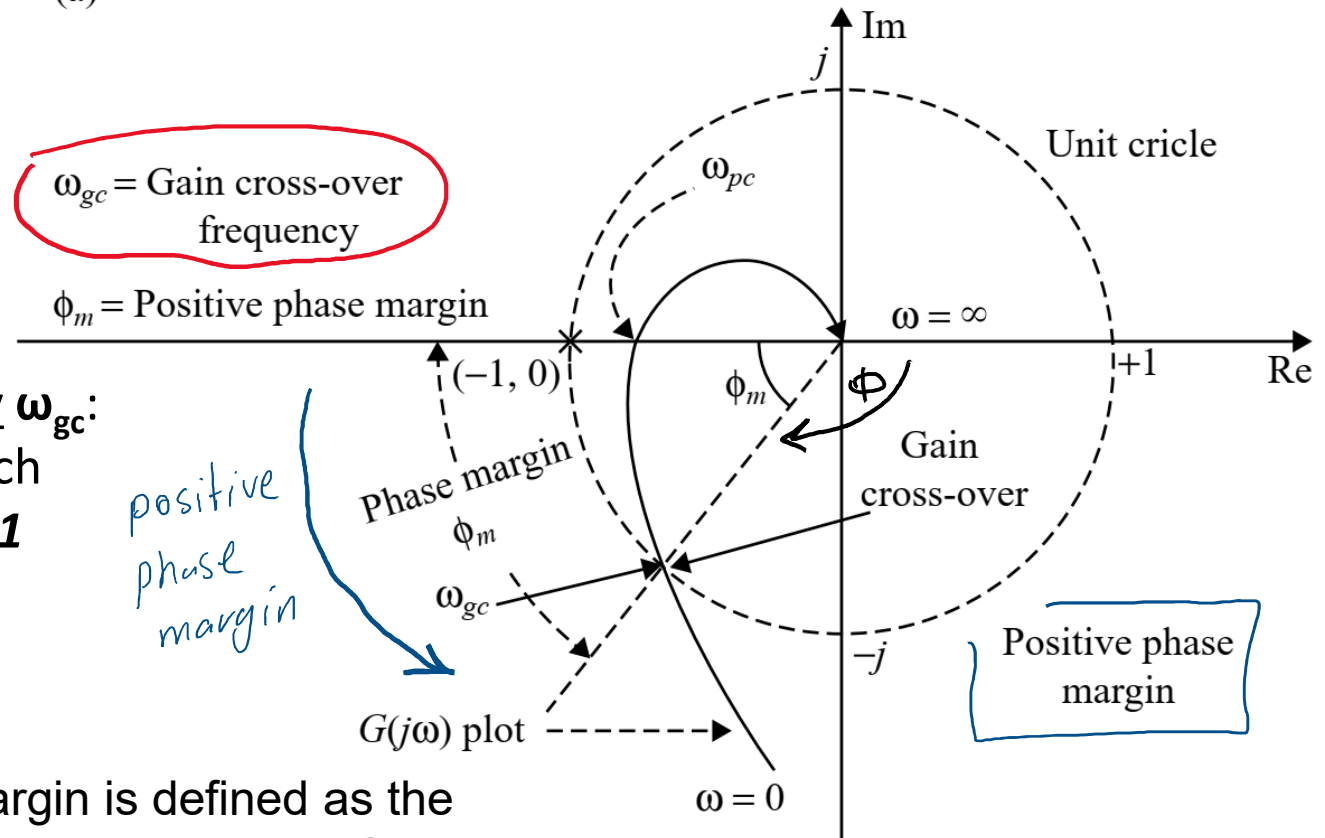


If the $G(j\omega)$ plot passes through $(-1, 0)$ point, i.e., $G_m = 1$ or $G_m = 0$ dB, The closed-loop system is oscillatory.

If the $G(j\omega)$ plot does not intersect the negative real axis (system of order two and less) then $G_m = \infty$

Phase margin (using Nyquist/Polar plot)

(a)



Gain crossover frequency ω_{gc} :

It is the frequency at which
the gain of $\mathbf{G}(j\omega)|_{\omega=\omega_{gc}} = 1$

phase difference before
instability.

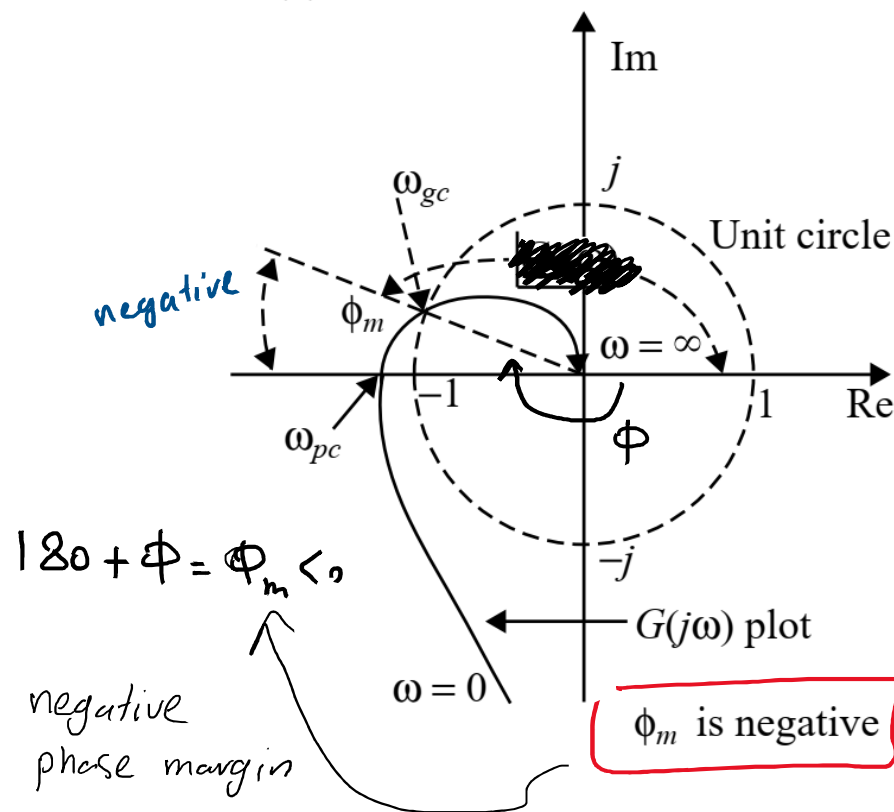
Phase Margin: Phase margin is defined as the amount of phase angle through which the $G(j\omega)$ plot should be rotated about the origin so that the gain crossover passes through the $(-1, 0)$ point. Phase margin is also defined as:

$$180 + \underbrace{\text{the phase angle of } G(j\omega) \text{ at unity gain}}_{\Phi} = \Phi_m \quad (\text{phase margin})$$

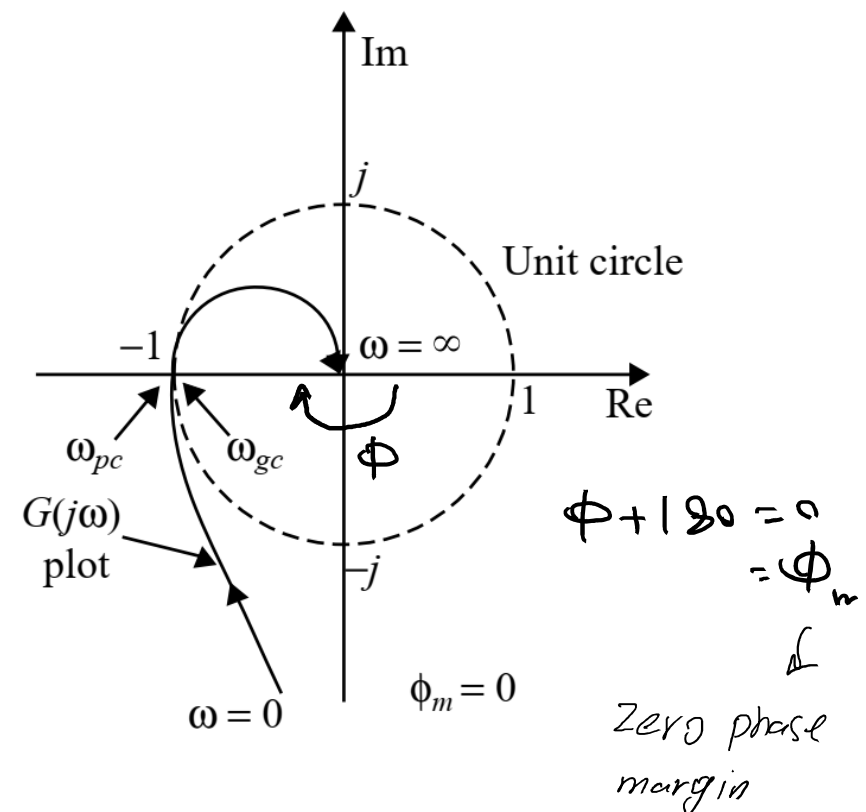
**For stability of the closed-loop system, both phase and gain margins should be positive!
(Necessary and Sufficient)**

Phase margin (using Nyquist/Polar plot) cont.

(b)



(c)



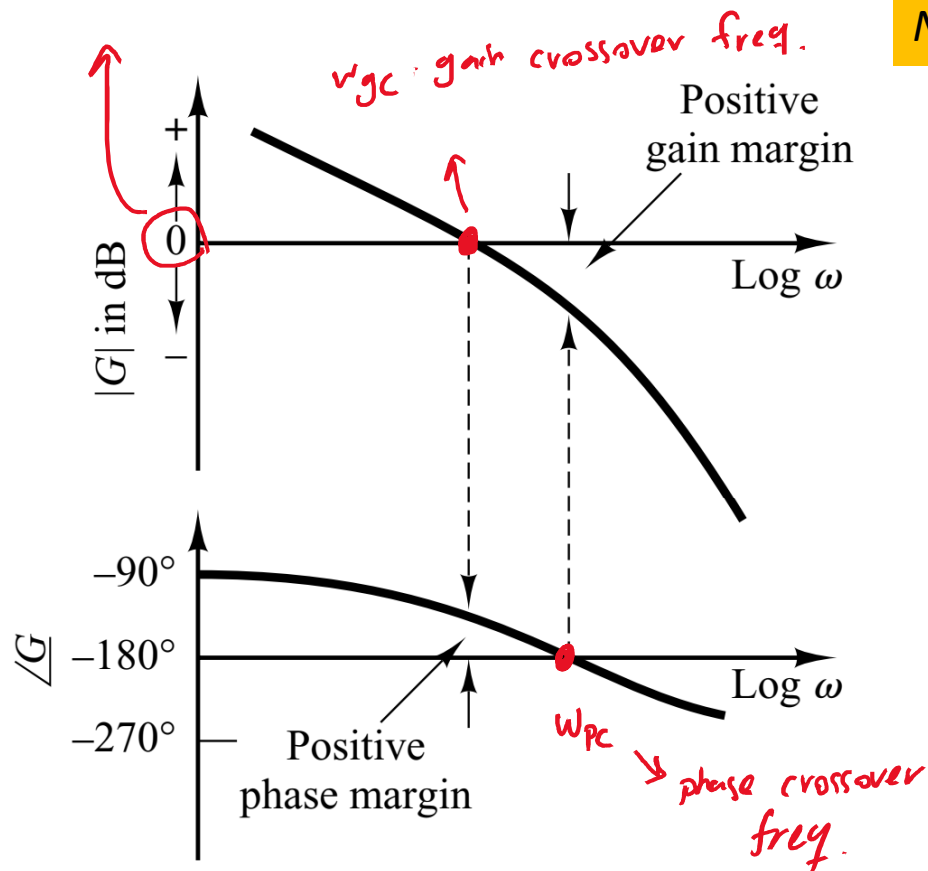
If the gain crossover lies in the third quadrant of the complex $G(j\omega)$ plane the **phase margin is positive**.

✓✓ For the system with a positive phase margin, an additional (negative) phase can be included in the open-loop system before the closed-loop system becomes unstable.

Gain and Phase Margin in Bode Plot

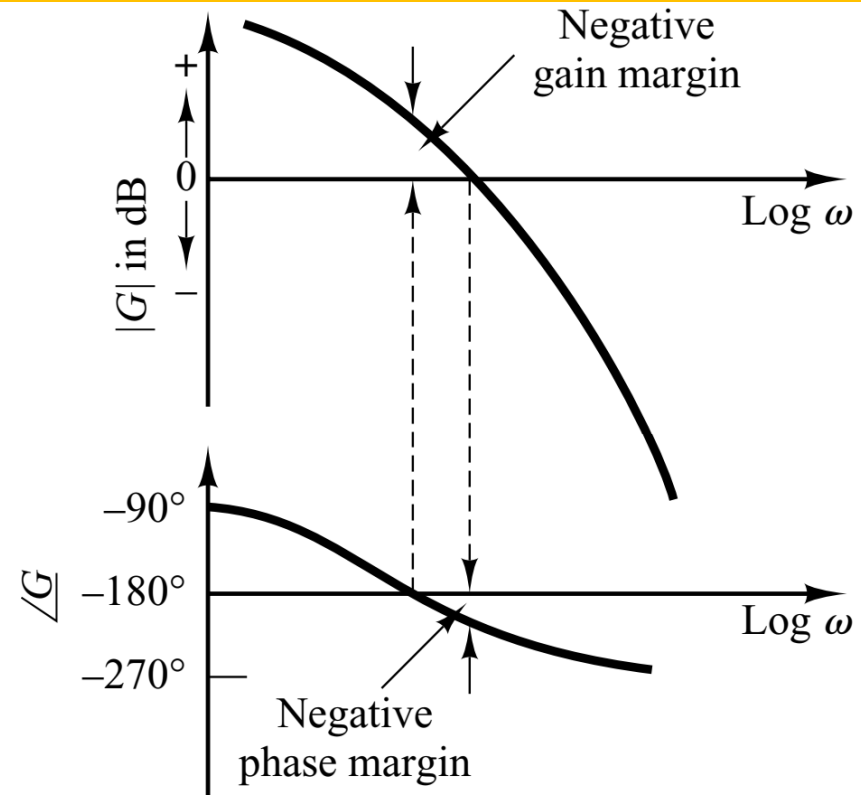
$$20 \log |G(j\omega)| = 0 \Rightarrow |G(j\omega)| = 1$$

Note: In discussing about stability margins we use the **Open-loop transfer function's Bode and Nyquist Plots** (not the closed-loop one!)



Stable system

Both phase and gain margins are positive.



Unstable system

both phase and gain margins are negative.

References

- N. Nise, Control Systems Engineering, 8th edition, John Wiley & Sons, 2020.
- S. Palani, Automatic Control Systems With MATLAB, 2nd edition, Springer, 2020.
- B. Dorf, R. Bishop, Modern Control Systems, 14th edition, Pearson, 2022.
- K. Ogata, Modern Control Engineering, 5th edition, Prentice Hall, 2010

Useful Videos on Bode Plots (Optional):



You can also watch **Brian Douglas'** videos on Bode Plots here:



Bode Plots and Frequency Response



Gain and phase margins